Graphing Rational Functions

Name _____

Algebra 2 Lesson 4-2 (fast track) Date _____ Period _____

Learning goal: I can find the domain and the vertical and horizontal asymptotes of a rational function from its equation.

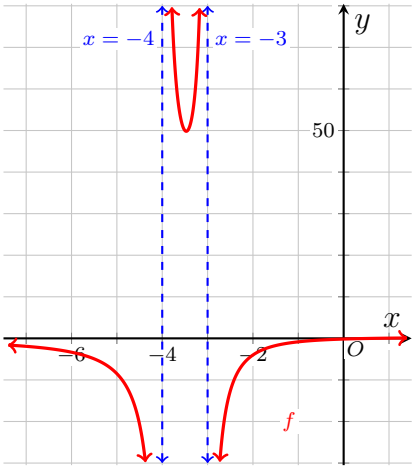
Language goal: I can explain how I found each asymptote using the words numerator, denominator, degree, asymptote, and leading coefficient.

On the quiz: (1) Match a graph to its equation (full Savvas quiz only). (2) Bonus, +5: find the vertical asymptotes when a factor of the denominator cancels with the numerator (Example 2). (3) Graded, 10 points: find both asymptotes of a linear-over-linear function (Examples 2 and 3). (4) Water added to a mixture (full Savvas quiz only); the Practice block is optional preparation for it. (5) Graded, 10 points: pick the function with a given horizontal asymptote by comparing degrees and leading coefficients (Example 3).

On the Topic 4 test: Question 2 asks for a domain (Example 2). Questions 3 and 12 ask for both asymptotes when a factor cancels (Examples 2, 3). Question 18 asks which functions have horizontal asymptote $y = 7$ (Example 3). Questions 8 (a remainder) and 16 (asymptotes of $\frac{ax+b}{cx+d}$) come from Examples 1 and 4, which this packet skips.

Example 2: Find Multiple Vertical Asymptotes of a Rational Function

What are the vertical asymptotes for the graph of $f(x) = \dfrac{3x-2}{x^2+7x+12}$?



Vertical **asymptotes** can occur at the x-values where the function is undefined. Determine where the **denominator** of the rational function is equal to 0.

Step 1 Set the denominator equal to zero and solve.

$x^2 + 7x + 12 = 0$

$(x+3)(x+4) = 0$

$x + 3 = 0$ or $x + 4 = 0$

$x = -3$ or $x = -4$

The possible vertical asymptotes are $x = -3$ and $x = -4$.

Step 2 Check the **numerator** there: $3(-3) - 2 = -11$ and $3(-4) - 2 = -14$. Neither is 0, so both are asymptotes.

Graph the function to check: the graph is not defined at $x = -3$ or $x = -4$. **These lines are vertical asymptotes.**

Rule (Concept Summary): $x = a$ is a vertical asymptote of $\frac{P(x)}{Q(x)}$ if $Q(a) = 0$ and $P(a) \neq 0$.

Cancel all common factors first. If $x = a$ still makes the reduced denominator 0, it is a vertical asymptote; if not, the graph has a hole at $x = a$. Either way, a is not in the domain.

Domain (the x-values where $Q(x) \neq 0$): all real numbers except -3 and -4.

Say it: The vertical asymptotes are $x =$ _____ and $x =$ _____ because the _____ is 0 there and the _____ is not.

Try It 2: Vertical asymptotes

Find the vertical asymptotes for each function. Graph the function to check your work.

a. $g(x) = \dfrac{5x}{x^2 - x - 6}$

denominator factored: _____

denominator is 0 at: _____

numerator at those x-values: _____

vertical asymptotes: _____

b. $h(x) = \dfrac{7 - x}{(x - 5)(x + 1)(x + 3)}$

denominator is 0 at: _____

numerator at those x-values: _____

vertical asymptotes: _____

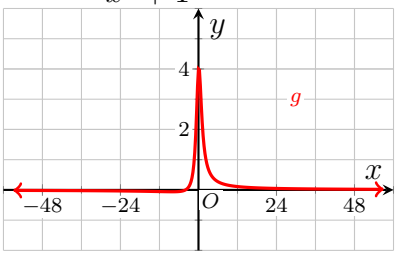
Example 3: Find Types of Horizontal Asymptotes

What are the horizontal asymptotes for the graph of $f(x) = \dfrac{3x - 2}{x^2 + 7x + 12}$?

A horizontal asymptote describes **end behavior**: what y does as $x \to \infty$ or $x \to -\infty$. Compare the **degree** (highest power) of the **numerator** with the degree of the **denominator**. There are three cases.

Case 1: degree of numerator < degree of denominator. Horizontal asymptote $y = 0$.

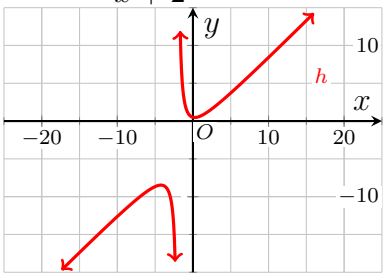
$g(x) = \dfrac{x + 4}{x^2 + 1}$ $\quad (1 < 2)$



The bottom grows faster, so $g(x) \to 0$.

Case 2: degree of numerator > degree of denominator. **No** horizontal asymptote.

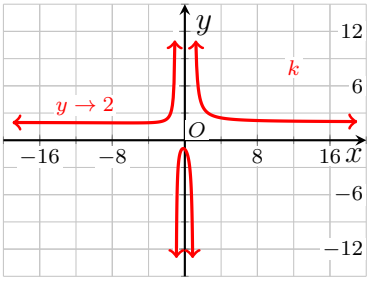
$h(x) = \dfrac{x^2 + 1}{x + 2}$ $\quad (2 > 1)$



The top grows faster, so $y \to \pm\infty$.

Case 3: degrees equal. Horizontal asymptote $y =$ the ratio of the **leading coefficients**.

$k(x) = \dfrac{2x^2 + x + 1}{x^2 - 1}$ $\quad (2 = 2)$



Long division:

$k(x) = 2 + \dfrac{x + 3}{x^2 - 1}$,

so $y = \frac{2}{1} = 2$.

Back to f. $f(x) = \dfrac{3x - 2}{x^2 + 7x + 12}$: the numerator has degree 1 and the denominator has degree 2. That is Case 1, so the horizontal asymptote is $\boldsymbol{y = 0}$.

Say it: The degree of the numerator of f is _____ and the degree of the denominator is _____, so the horizontal asymptote is _____ because ...

Try It 3: Horizontal asymptotes

What are the horizontal asymptotes of the graph of each function?

a. $g(x) = \dfrac{2x^2 + x - 9}{2x - 8}$

degree on top: _____

degree on bottom: _____

horizontal asymptote:

_____

b. $h(x) = \dfrac{x^2 + 5x + 4}{3x^2 - 12}$

degree on top: _____

degree on bottom: _____

horizontal asymptote:

_____

c. $k(x) = \dfrac{x}{(2x - 1)(x + 6)}$

degree on top: _____

degree on bottom: _____

horizontal asymptote:

_____

Say it: In part b the asymptote is not $y = 0$ because the degrees are _____, so I use the ratio of the _____.

Practice: Adding water to a mixture

A juice maker adds x liters of water to a tank that holds 5 liters of a drink that is 20% sugar. The fraction of the new mixture that is sugar is $f(x) = \dfrac{1}{x + 5}$.

a. Why is the numerator 1?

_____

_____

b. How many liters of water must be added to make a 4% sugar mixture? (4% = 0.04. Solve $\frac{1}{x+5} = 0.04$, or graph $y = f(x)$ and $y = 0.04$ and find where they meet.)

_____

_____

_____

c. What is the horizontal asymptote of f? What does it mean for the drink when a lot of water is added?

_____

_____

_____

d. The vertical asymptote is $x = -5$. Why does it not matter here?

_____

_____

IXL: Practice for the quiz

Go to IXL, Algebra 2. Type the code in the search box. Do the starred skills first.

Code	Skill	Feeds
7JJ ★	Rational functions: asymptotes and excluded values	Quiz 2, 3 and 5: both asymptotes from the equation (Examples 2 and 3)
79Q ★	Domain of rational functions	Quiz 2 and 3: the denominator is 0 at a vertical asymptote and the value is left out of the domain (Example 2)
ULD	Domain and range of reciprocal functions	Warm-up from Lesson 4-1 (Quiz 1, full Savvas quiz only)
UB5 ★	Factor quadratics	Quiz 2 (bonus): factor a denominator to find where it is 0 (Example 2, Try It 2a, Exit question)

SmartScore 70 or above on each starred (★) skill replaces your quiz score.

Exit question: Describe and correct the error

Juanita is trying to determine the vertical and horizontal asymptotes for the graph of the function $f(x) = \dfrac{x^2 + 3x - 4}{x^2 - x - 12}$. Here is her work:

$$f(x) = \frac{x^2 + 3x - 4}{x^2 - x - 12} = \frac{(x+4)(x-1)}{(x+3)(x-4)}$$

vertical asymptote: $x = -3$, $x = 4$

horizontal asymptote: $y = -4$, $y = 1$ ×

Describe and correct the error Juanita made in determining the vertical and horizontal asymptotes.

Juanita's vertical asymptotes are _____ because the _____ is 0 at $x = -3$ and $x = 4$ and the numerator is not.

Her horizontal asymptotes are wrong because she used _____.

The degree of the numerator is _____ the degree of the denominator, so the horizontal asymptote is $y =$ _____, the ratio of the _____.